Bayes' Theorem

Lecture/Lab 20

Physics 4730, Fall 2026



Thomas Bayes,
(1701-1761)

Remainder of Course

- Bayes' Theorem (today)
- Machine Learning (starting Tuesday 11/10)
- Final Project (introduced Tuesday 11/3)

Classical vs Bayesian Inference

Classical (Frequentist)

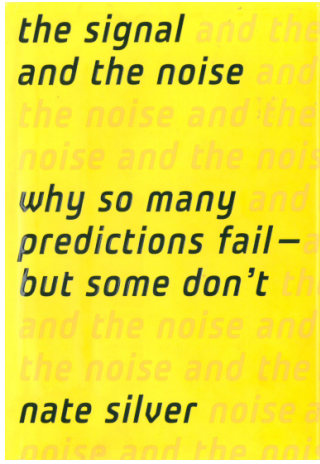
- What we've done so far
- probabilities refer to relative frequencies of events
- parameters are fixed constants

Bayesian

- Bayes 1700's, Laplace
- probabilities are statements of subjective belief
- parameters are treated like RVs

Bayes Canonical Example

- A deadly disease afflicts 0.1% of the population.
- A test exists that detects the disease 99% of the time, and only gives false positives 1% of the time.
- **You** test positive. How worried should you be?



Bayes' Theorem

Let E (event) be that you tested positive
Let H (hypothesis) be that you have the disease

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

where

$P(H|E) \equiv$ probability you have disease, given you tested positive ("posterior")

$P(E|H) \equiv$ probability you tested positive, given you have disease

$P(H) \equiv$ probability that you have disease ("prior")

$P(E) \equiv$ total probability of testing positive (normalization)

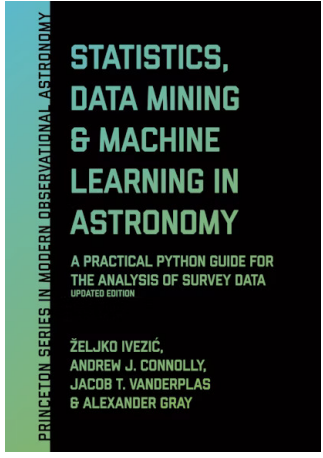
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$$
\begin{aligned}
P(H) &= 0.001 \\
P(E|H) &= 0.99 \\
P(E) &= P(H) \times P(E|H) \\
&\quad + P(-H) \times P(E|-H) \\
&= 0.001 \times 0.99 + 0.999 \times 0.01 \\
&= 0.01098 \\
\therefore P(H|E) &= 0.092 \rightarrow \text{you are likely ok!}
\end{aligned}
$$

Proof of Bayes' Theorem from Kolmogorov Axioms

See, for example:



Example (Today's Lab): Monte Hall Problem



Monty Hall (1921–2017), host of "Let's Make a Deal"

The Game:

- A prize is hidden behind one of three doors
- Choose a door (1, 2, or 3) at random.
- Monty shows you one of the remaining doors is empty, and offers you a chance to switch your choice.
- Should you switch?

Monty Hall Problem: Solution Using Bayes' Theorem

Let H_i be hypothesis prize is behind door i (of N)

$$p(H_i) = \frac{1}{N} \qquad \text{the prior}$$

Player chooses door $i = 1$ (without loss of generality). Then, let

$$d_j, \quad j \neq 1$$

be the data that door j remains closed.

Now, apply Bayes' Rule. The probability that the prize is behind door 1, given that door j remains closed, is:

$$P(H_1|d_j) = \frac{P(d_j|H_1) \times P(H_1)}{P(d_j)}$$

The probability j is unopened given H_1 is

$$p(d_j|H_1) = \frac{1}{N-1}$$

The probability the prize is behind door 1

$$p(H_1) = \frac{1}{N}$$

The probability door j is unopened is

$$\begin{aligned} p(d_j) &= \sum_{i=1}^{N} p(d_j|H_i)p(H_i) \\ &= p(d_j|H_1)p(H_1) + \sum_{i=2}^{N} p(d_j|H_i)p(H_i) \\ &= p(d_j|H_1)p(H_1) + p(d_j|H_j)p(H_j) \\ &= \frac{1}{N-1} \times \frac{1}{N} + 1 \times \frac{1}{N} \\ &= \frac{1}{N-1} \end{aligned}$$

and

$$p(H_1|d_j) = \frac{\left(\frac{1}{N-1}\right)\left(\frac{1}{N}\right)}{\left(\frac{1}{N-1}\right)} = \frac{1}{N}$$

Now, the probability that the prize is behind door j, given that door j remains closed, is:

$$\begin{aligned} P(H_j|d_j) &= \frac{P(d_j|H_j) \times P(H_j)}{P(d_j)} \\ &= \frac{1 \times \frac{1}{N}}{\left(\frac{1}{N-1}\right)} \\ &= 1 - \frac{1}{N} \end{aligned}$$

which is greater than

$$p(H_1|d_j) = \frac{1}{N}$$

so you should switch!

Lab 20

- Using random numbers, check "experimentally" the Bayesian Inference of the Monty Hall problem worked out in lecture.
- Potentially useful: **numpy.random.randint**
- *For an added challenge*: Conduct the experiment for an arbitrary number N of doors, of which $N-2$ empty doors are revealed following the contestant's initial guess.